Cpts 317. 12/7

HW Hint.

Prob 1. (1). Since L is r.e., we have a TM M s.t.

$\forall x$,

$x \in L \Rightarrow M$ says yes on x;

$x \notin L \Rightarrow M$ doesn't say yes on x.

Since $\bar{L}$ is r.e., we have a TM $\bar{M}$ s.t.

$\forall x$,

$x \in \bar{L} \Rightarrow \bar{M}$ says yes on x;

$x \notin \bar{L} \Rightarrow \bar{M}$ doesn't say yes on x.

Recall: $x \in \bar{L}$ iff $x \notin L$;

$x \notin \bar{L}$ iff $x \in L$.

To show L is recurse, we need construct an alg $\hat{M}$ to recog L:

input x

Run M on x and

Run $\bar{M}$ on x, in parallel.

if M says yes on x, ret yes

if $\bar{M}$ says yes on x, ret No.

Prob 1 (2). We only need to show that there is a lang L s.t. L is r.e. but $\bar{L}$ is not. (many ways).

Take $L = L_{self}$. We know L_{self} is r.e.,

but $L_{\neg self} = \bar{L}$ is not r.e..

Prob 2. We need show: if L is recursive, then so is $\bar{L}$.

By def, L is recursive if there is an alg M to recog L:

$\forall x$,

$x \in L \Rightarrow M$ says yes on x,

$x \notin L \Rightarrow M$ says no on x.

To show $\bar{L}$ is recursive, we construct an alg $\bar{M}$ to recog it:

input x

Run M on x

if M says yes on x, ret NO;

if M says no on x, ret yes.

$\blacksquare$.

Prob3. Need show: for each M, there are inf. many M_i with $L(M) = L(M_i)$.

(add garbage instructions: work on it.)

Prob4. my alg:

input $\langle M \rangle$ // code of TM M.

I count in the code of M, the # of L-inst, and the # of R-inst, and see if they are the same.

// Do I need to run
// M in order to answer
// the question? NO.

Basics:

B9. def's on recursiveness, r.e., decidable, undecidable

more ex:

Show: if L_1 is recursive, and L_2 is recursive,

Then so is $L_1 \cap \overline{L_2}$.

Proof. Since L_1 is recursive, we have an alg M_1 to recog it. $\forall x$,

$x \in L_1 \Rightarrow M_1$ says yes on x;

$x \notin L_1 \Rightarrow M_1$ says no on x.

Since L_2 is recursive, we have an alg M_2 to recog it. $\forall x$,

$$x \in L_2 \Rightarrow M_2 \text{ says yes on } x$$

$$x \notin L_2 \Rightarrow M_2 \text{ says no on } x.$$

We now construct an alg M to recog $L_1 \cap \overline{L_2}$:

input x

Run M_1 on x;

Run M_2 on x;

if M_1 says yes and M_2 says no on x,

then ret yes,

else ret No.

M: input x

Run M_1 on x and Run M_2 on x, in parallel.

if one of them says yes, ret yes.

Can you give me precisely an undecidable problem?

Given: a TM M with null input,

Qutn: ~~Do you~~ Does M run forever?

The following problem is decidable or not? prove it.

Given: a FA M

Question: Do M halt with input 'abb'?

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yes, decidable!

Show the following is decidable:

Given: a PDA M

Question: Do M say yes on 'abb'?

// Hint: M can run forever on 'abb'.
//
//

Proof. We do alg to solve emptiness of PDA:

Given: PDA M

Qsth: $L(M) = \emptyset$?

From the given M, we construct a PDA M' s.t.

$$L(M') = \underbrace{L(M)}_{\text{c.f.}} \cap \underbrace{\{'abb'\}}_{\text{reg.}}.$$

Then the result follows since we have alg to check "$L(M') = \emptyset$?".